Stress Analysis Report

Analyzed File:	1426_Mitoiu.Bianca-Maria_Ansamblu.iam
Autodesk Inventor Version:	2020 (Build 240168000, 168)
Creation Date:	1/6/2020, 12:09 AM
Study Author:	Mitoiu Bianca-Maria
Summary:	

Project Info (iProperties)

Summary

Author	Bianca

Project

Part Number	1426_Mitoiu.Bianca-Maria_Ansamblu
Designer	Bianca
Cost	$0.00
Date Created	1/5/2020

Status

Design Status	WorkInProgress

Physical

Mass	0.0295692 lbmass
Area	17.3829 in^2
Volume	0.227397 in^3
Center of Gravity	x=-2.88515 in y=0.163668 in z=-2.04834 in

Note: Physical values could be different from Physical values used by FEA reported below.

Static Analysis:1

General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	1/6/2020, 12:08 AM
Detect and Eliminate Rigid Body Modes	No
Separate Stresses Across Contact Surfaces	No
Motion Loads Analysis	No

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	No
Use part based measure for Assembly mesh	Yes

Material(s)

Name	Aluminum 6061	
General	Mass Density	0.0975437 lbmass/in^3
	Yield Strength	39885.4 psi
	Ultimate Tensile Strength	44961.7 psi
Stress	Young's Modulus	0.00001 ksi
	Poisson's Ratio	0.33 ul
	Shear Modulus	0.0000037594 ksi
Part Name(s)	1426_Mitoiu.Bianca-Maria_Piesa1.ipt	

Name	Iron, Cast	
General	Mass Density	0.25831 lbmass/in^3
	Yield Strength	109939 psi
	Ultimate Tensile Strength	128213 psi
Stress	Young's Modulus	17477 ksi
	Poisson's Ratio	0.3 ul
	Shear Modulus	6721.94 ksi
Part Name(s)	1426_Mitoiu.Bianca-Maria_piesa2.ipt 1426_Mitoiu.Bianca-Maria_piesa2_MIR.ipt 1426_Mitoiu.Bianca-Maria_Piesa3.ipt 1426_Mitoiu.Bianca-Maria_Piesa3_MIR.ipt	

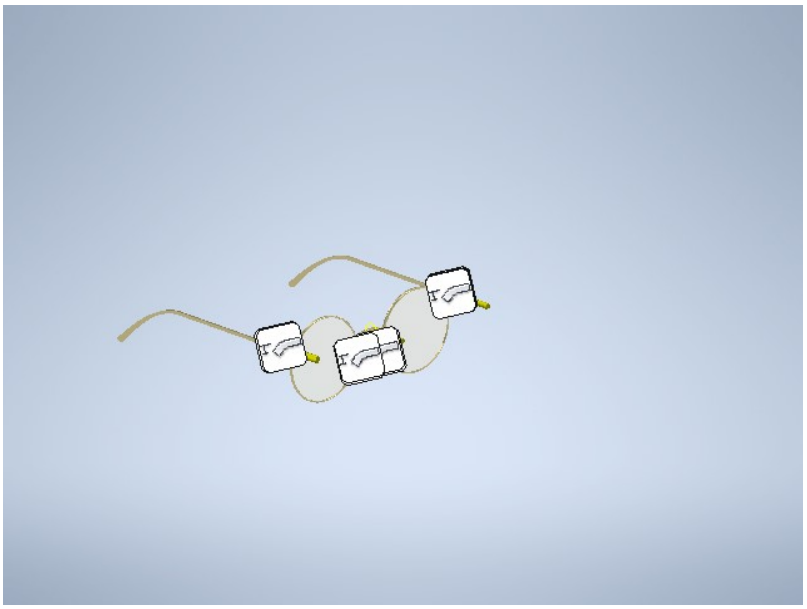
Name	Rubber, Silicone	
General	Mass Density	0.0451591 lbmass/in^3
	Yield Strength	1499.69 psi
	Ultimate Tensile Strength	942.745 psi
Stress	Young's Modulus	0.435113 ksi
	Poisson's Ratio	0.49 ul
	Shear Modulus	0.146011 ksi
Part Name(s)	1426_Mitoiu.Bianca-Maria_Piesa4.ipt 1426_Mitoiu.Bianca-Maria_Piesa4_MIR.ipt	

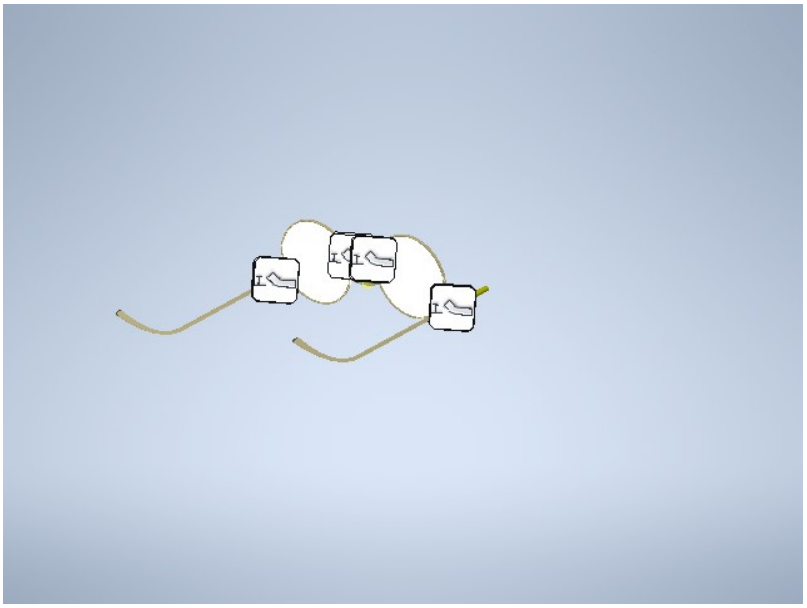
Operating conditions

Force:1

Load Type	Force
Magnitude	1.124 lbforce
Vector X	-1.027 lbforce
Vector Y	0.172 lbforce
Vector Z	-0.422 lbforce

Selected Face(s)

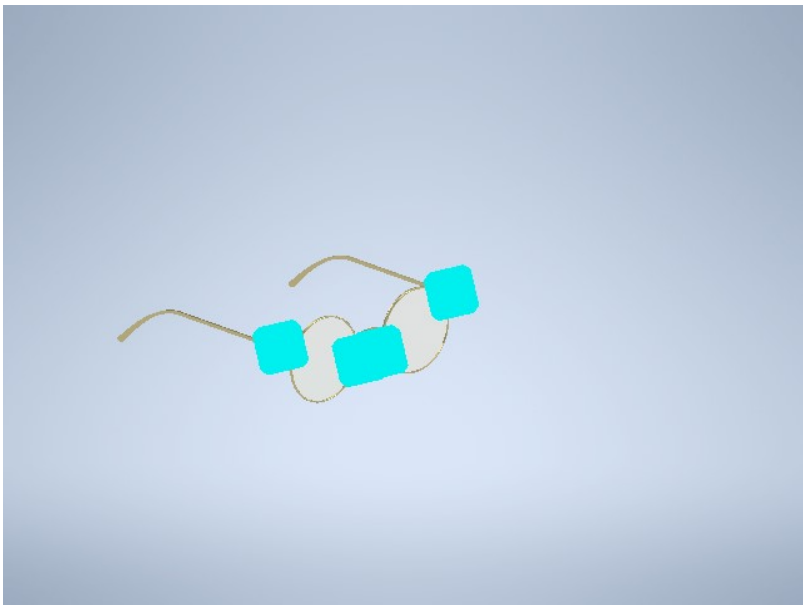


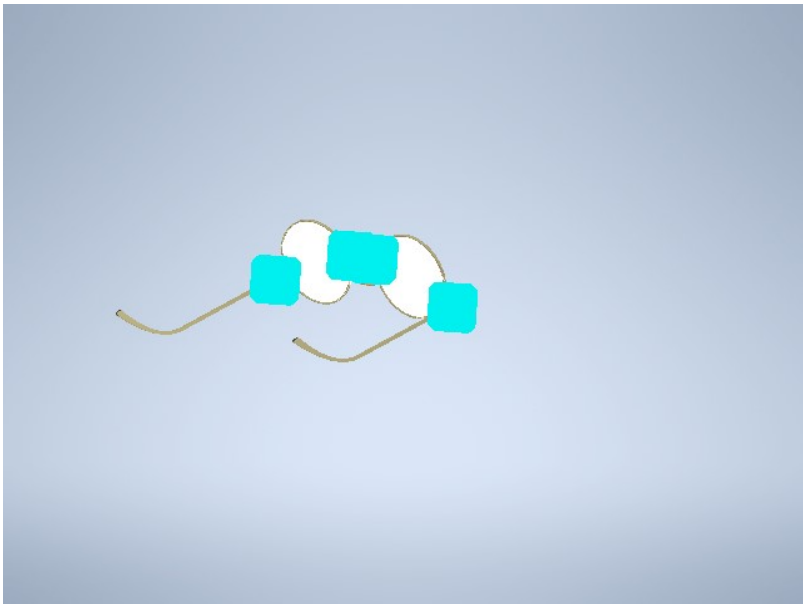


Fixed Constraint:1

Constraint Type	Fixed Constraint

Selected Face(s)





Contacts (Bonded)

Name	Part Name(s)
Bonded:1	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3:1
Bonded:2	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3:1
Bonded:3	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3_MIR:1
Bonded:4	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2:1
Bonded:5	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa1:1
Bonded:6	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2:1
Bonded:7	

	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3_MIR:1
Bonded:8	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3_MIR:1
Bonded:9	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2_MIR:1
Bonded:10	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2_MIR:1
Bonded:11	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3:1
Bonded:12	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa3:1
Bonded:13	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2:1
Bonded:14	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2:1
Bonded:15	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2:1
Bonded:16	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2_MIR:1
Bonded:17	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2_MIR:1
Bonded:18	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_piesa2_MIR:1
Bonded:19	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa1:1
Bonded:20	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa4_MIR:1
Bonded:21	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa4_MIR:1
Bonded:22	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa1:1
Bonded:23	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa4:1
Bonded:24	1426_Mitoiu.Bianca-Maria_Piesa1:1 1426_Mitoiu.Bianca-Maria_Piesa4:1
Bonded:25	1426_Mitoiu.Bianca-Maria_piesa2:1 1426_Mitoiu.Bianca-Maria_Piesa3_MIR:1
Bonded:26	1426_Mitoiu.Bianca-Maria_piesa2_MIR:1 1426_Mitoiu.Bianca-Maria_Piesa3:1

Results

Reaction Force and Moment on Constraints

Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	3.79411 lbforce	3.70045 lbforce	0.484483 lbforce ft	-0.158908 lbforce ft
		-0.713554 lbforce		-0.365788 lbforce ft
		-0.439012 lbforce		-0.275083 lbforce ft

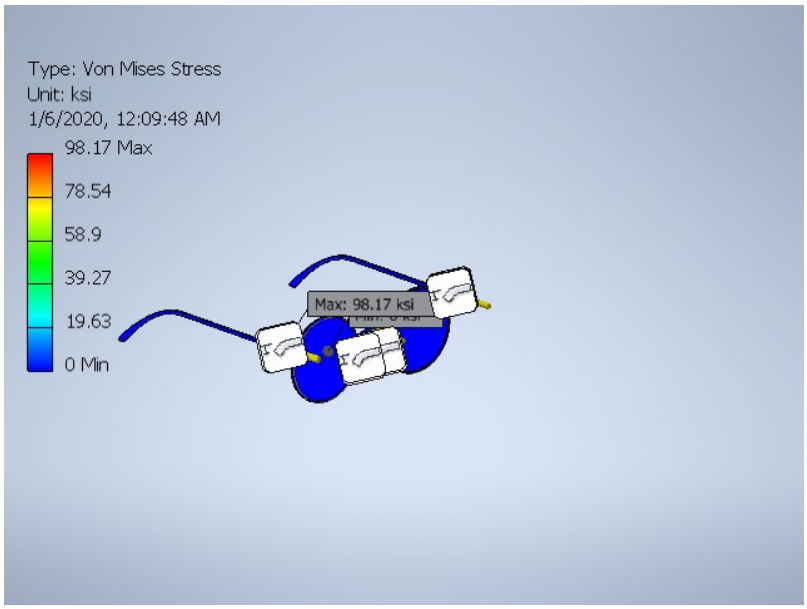
Result Summary

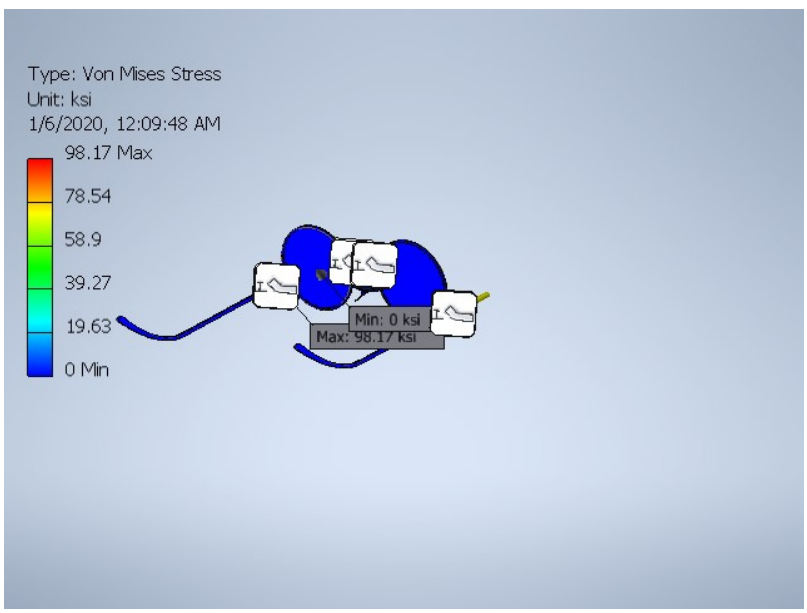
Name	Minimum	Maximum
Volume	0.227356 in^3	

Mass	0.0295652 lbmass	
Von Mises Stress	0 ksi	98.1697 ksi
1st Principal Stress	-17.2843 ksi	52.409 ksi
3rd Principal Stress	-123.599 ksi	7.25073 ksi
Displacement	0 in	2984.12 in
Safety Factor	1.11988 ul	15 ul
Stress XX	-110.859 ksi	51.4189 ksi
Stress XY	-8.88925 ksi	21.1332 ksi
Stress XZ	-27.0532 ksi	12.9427 ksi
Stress YY	-23.4948 ksi	13.4311 ksi
Stress YZ	-7.04785 ksi	10.6198 ksi
Stress ZZ	-42.9077 ksi	23.0905 ksi
X Displacement	-2653.29 in	153.018 in
Y Displacement	-401.385 in	988.377 in
Z Displacement	-1198.06 in	108.081 in
Equivalent Strain	0 ul	81948.4 ul
1st Principal Strain	-0.0201904 ul	48588.9 ul
3rd Principal Strain	-96426.7 ul	0.0754318 ul
Strain XX	-84009.8 ul	25626.7 ul
Strain XY	-44653.9 ul	40332.5 ul
Strain XZ	-30732.7 ul	31129.9 ul
Strain YY	-20036.9 ul	25396.7 ul
Strain YZ	-22785.3 ul	25621.3 ul
Strain ZZ	-23269.2 ul	17957.1 ul
Contact Pressure	0 ksi	273.578 ksi
Contact Pressure X	-174.277 ksi	229.343 ksi
Contact Pressure Y	-78.4718 ksi	65.1552 ksi
Contact Pressure Z	-104.734 ksi	140.69 ksi

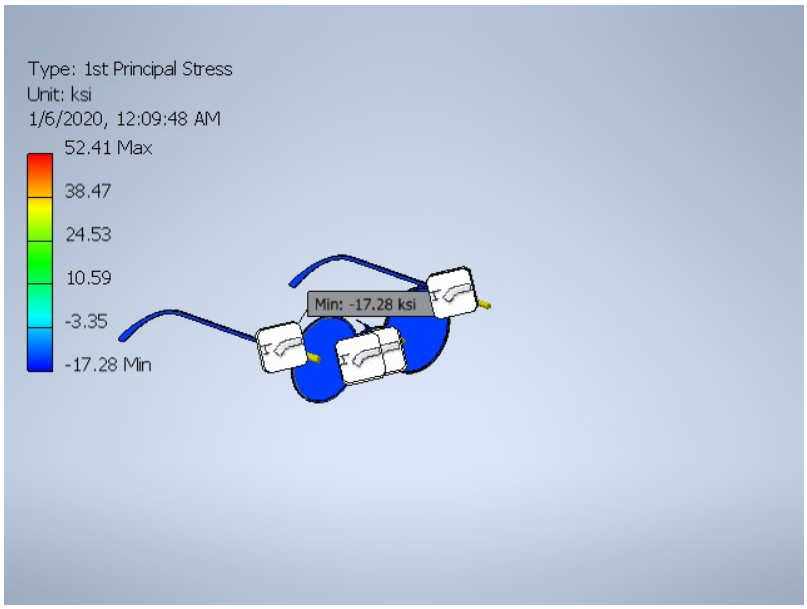
Figures

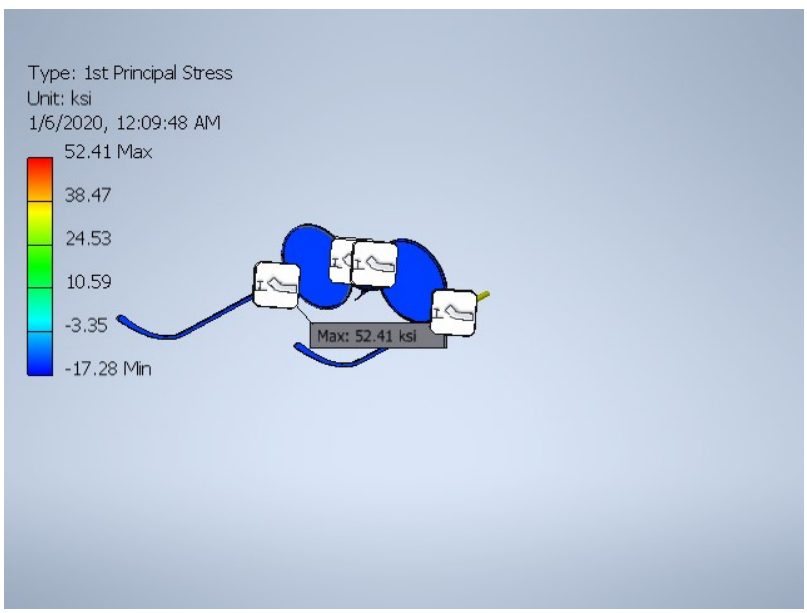
Von Mises Stress



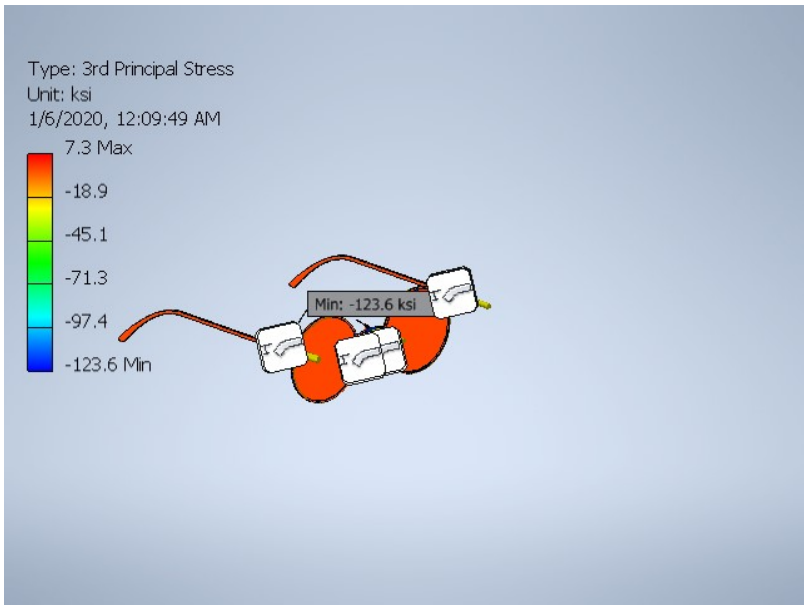


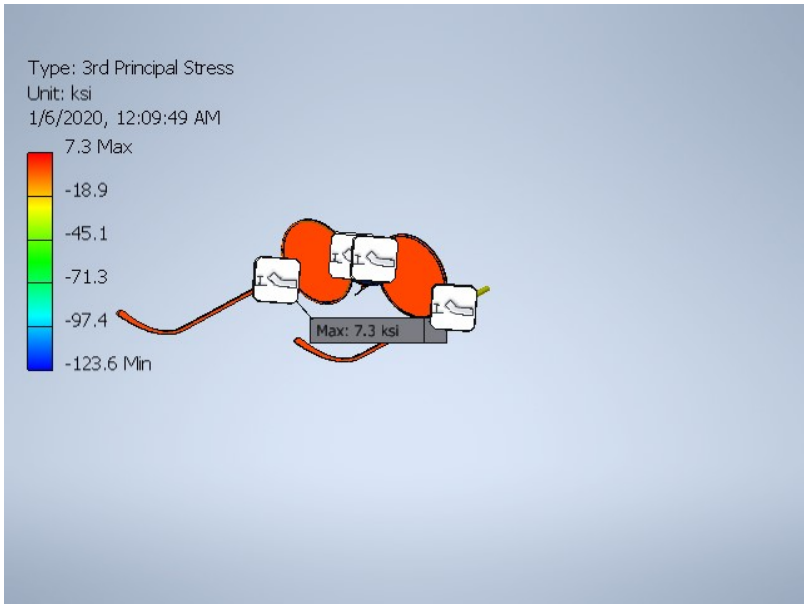
1st Principal Stress



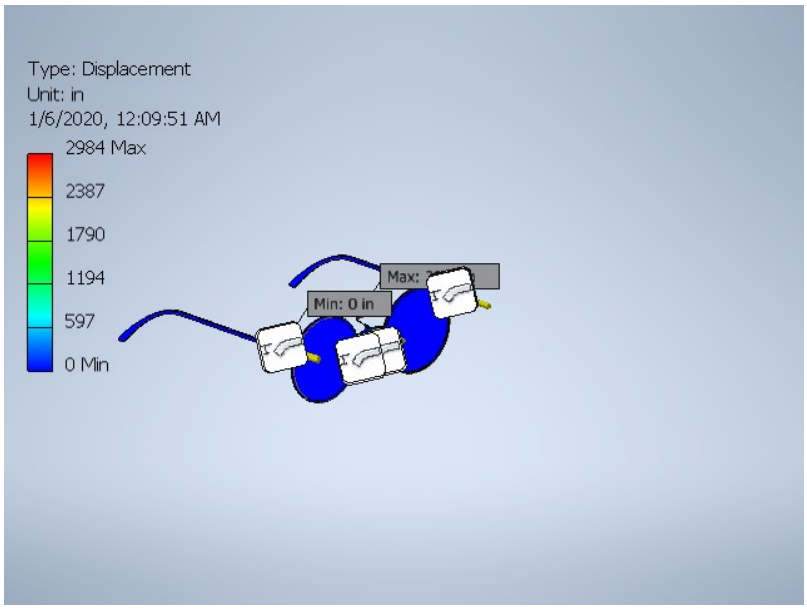


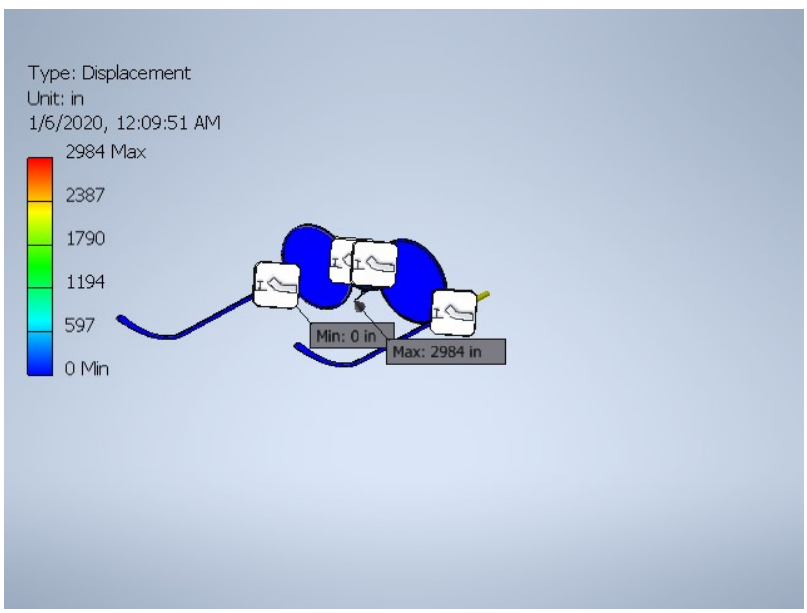
⊟ 3rd Principal Stress



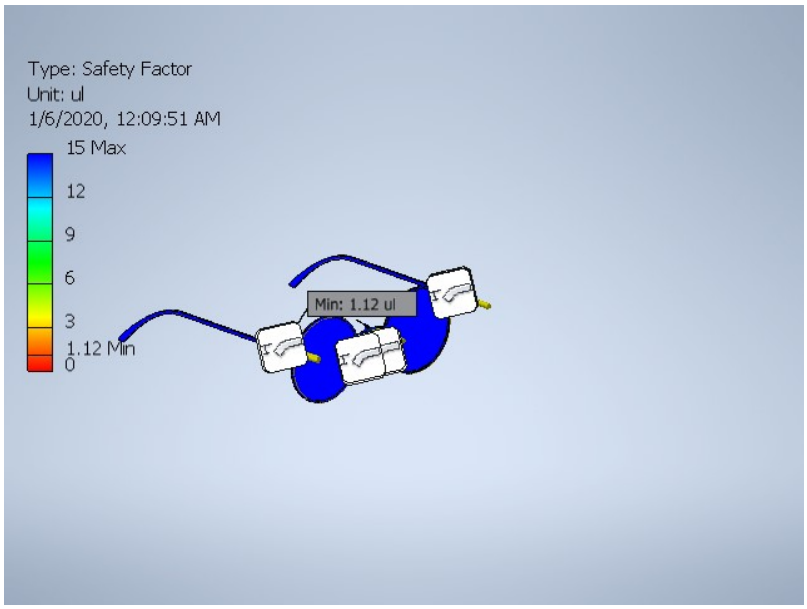


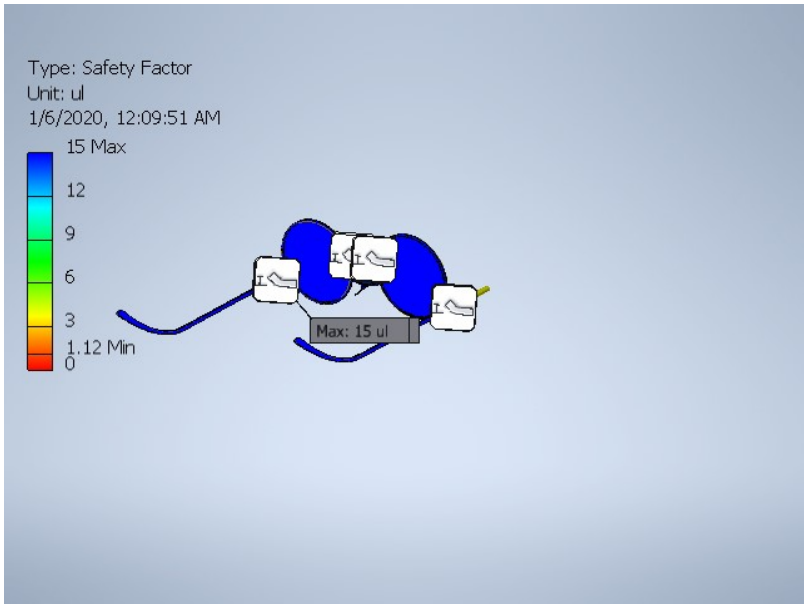
Displacement



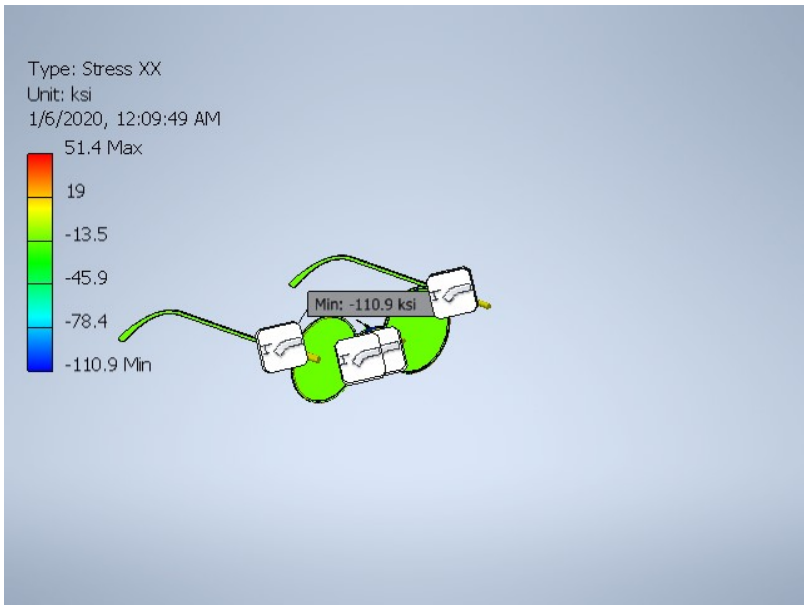


Safety Factor





Stress XX


Type: Stress XX
Unit: ksi
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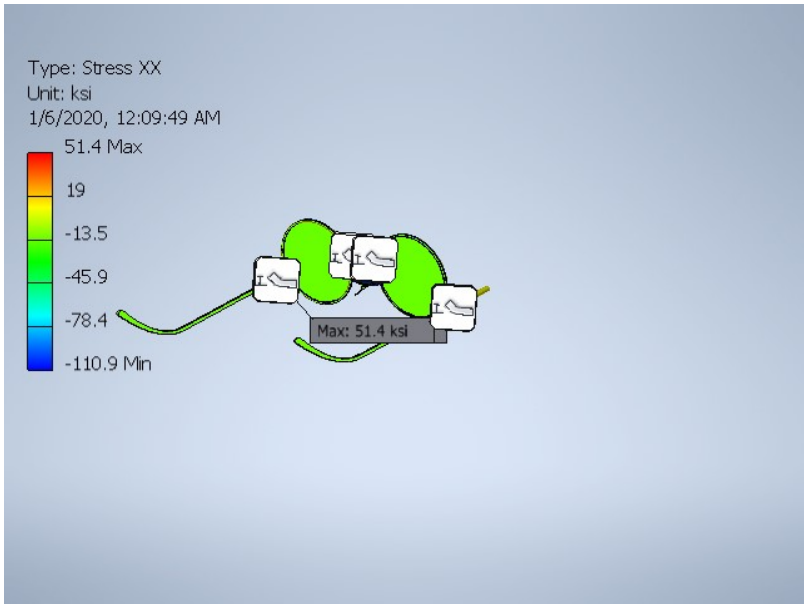
51.4 Max
19
-13.5
-45.9
-78.4
-110.9 Min
Min: -110.9 ksi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Unit: ksi
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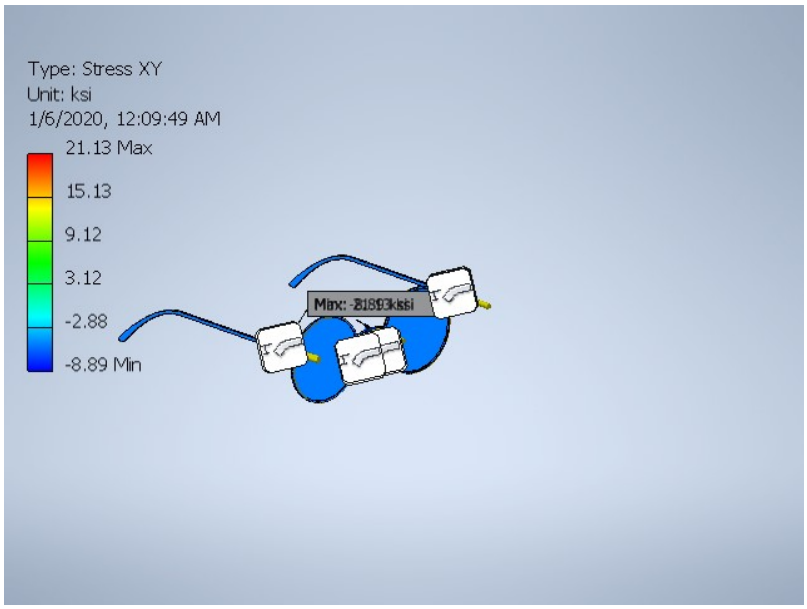
19
-13.5
-45.9
-78.4
-110.9 Min
Max: 51.4 ksi

Stress XY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Unit: ksi
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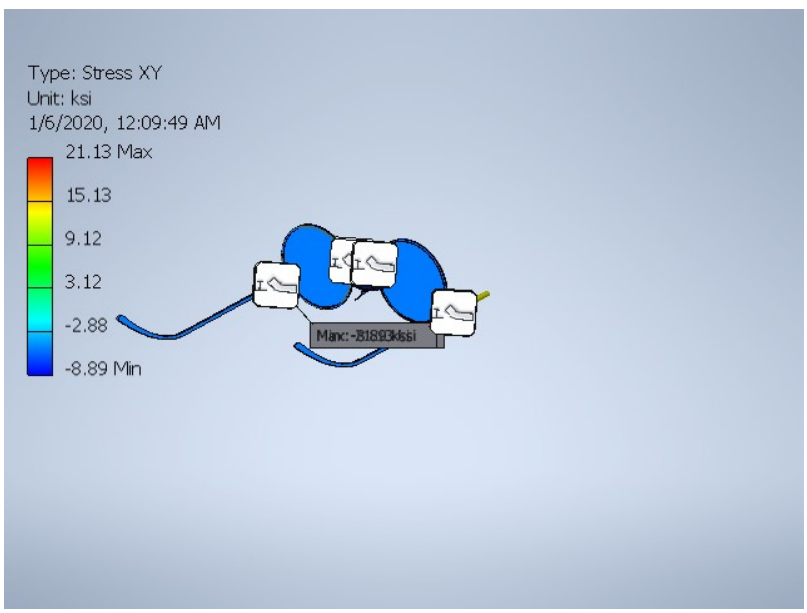
21.13 Max
15.13
9.12
3.12
-2.88
-8.89 Min
Max: -8.1893ksi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Unit: ksi
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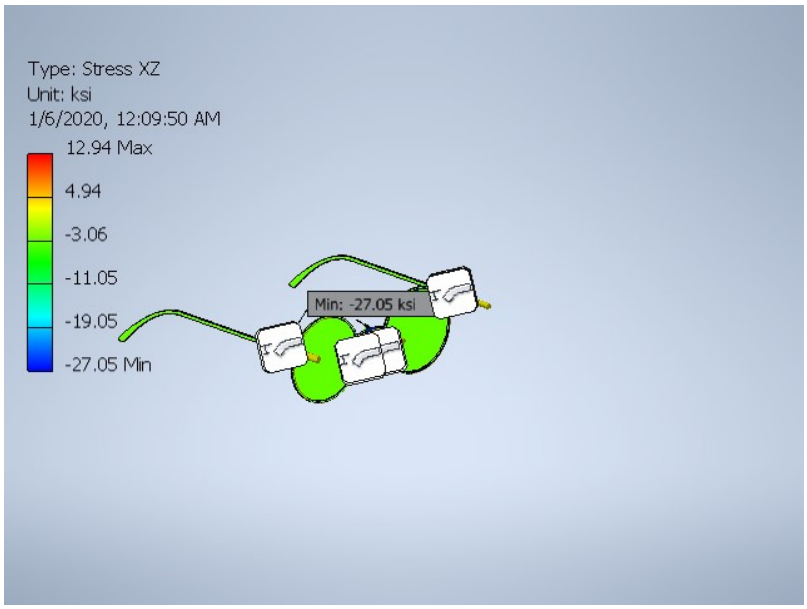
21.13 Max
15.13
9.12
3.12
-2.88
-8.89 Min
Max: -8.1893ksi

Stress XZ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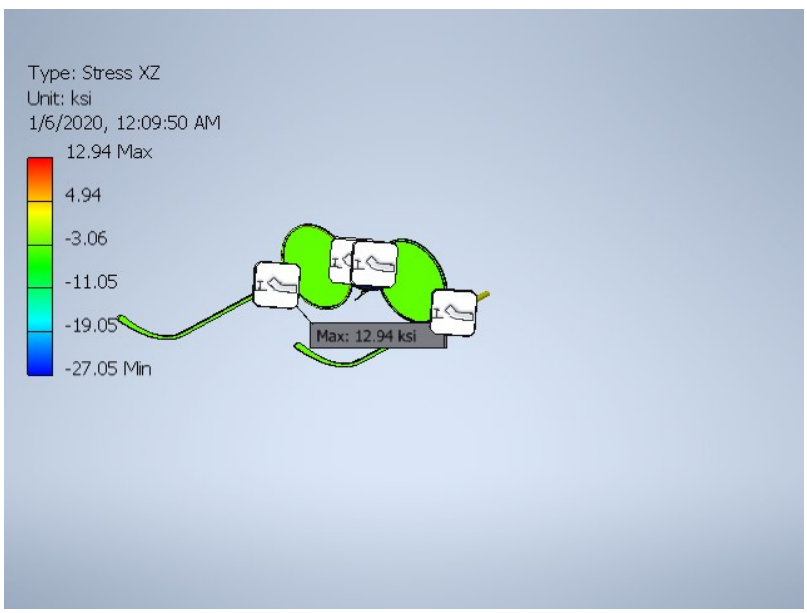
12.94 Max
4.94
-3.06
-11.05
-19.05
-27.05 Min
Min: -27.05 ksi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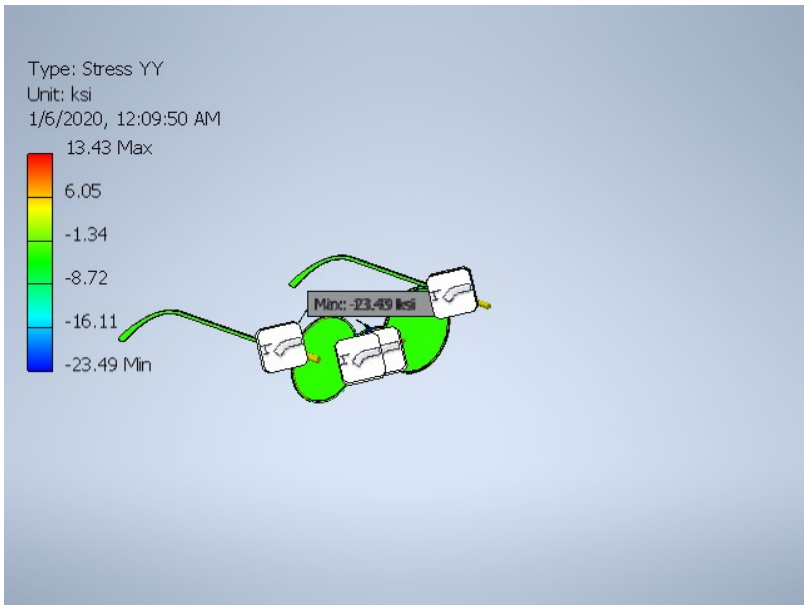
-3.06
-11.05
-19.05
-27.05 Min
Max: 12.94 ksi

Stress YY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13.43 Max
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-1.34
-8.72
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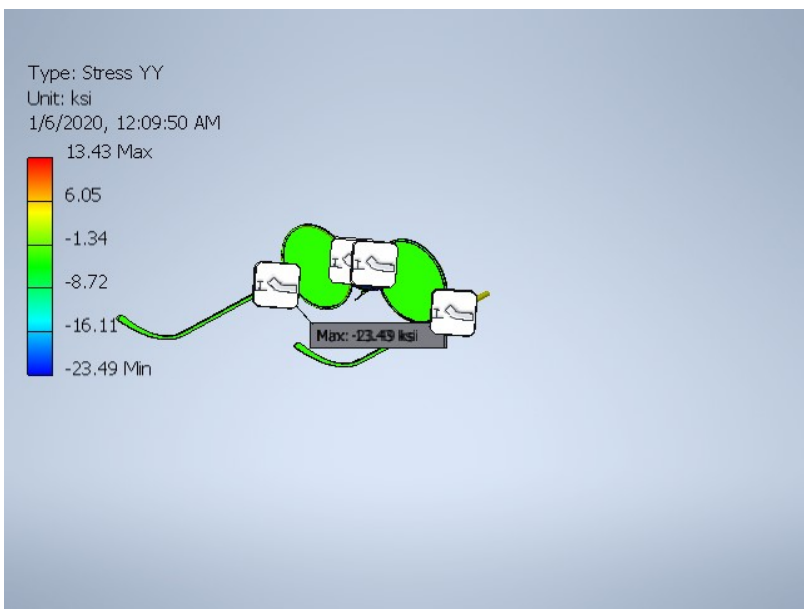
-23.49 Min
Min: -23.49 ksi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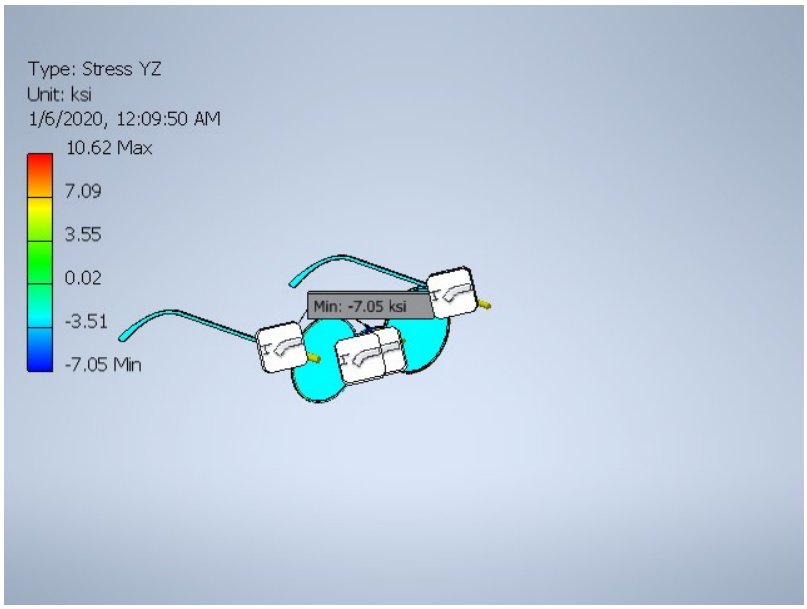
-8.72
-16.11
-23.49 Min
Max: -23.49 ksi

Stress YZ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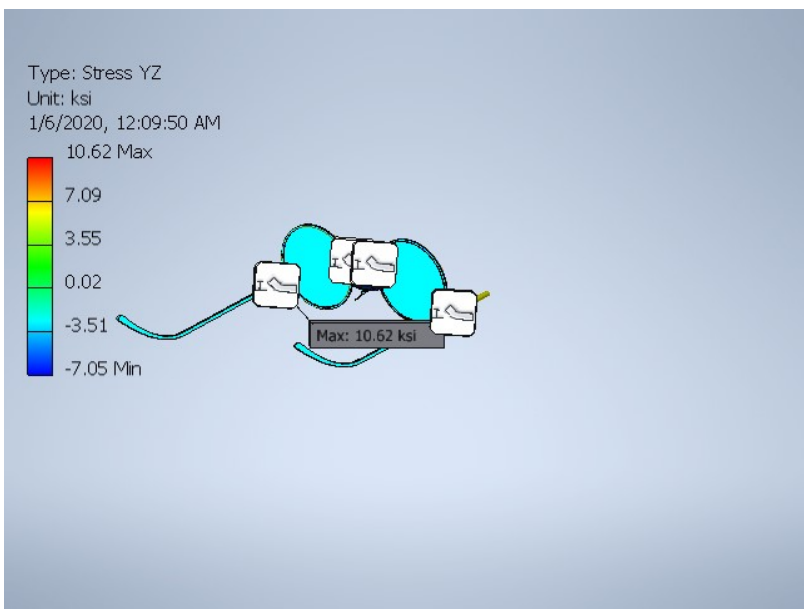
10.62 Max
7.09
3.55
0.02
-3.51
-7.05 Min
Min: -7.05 ksi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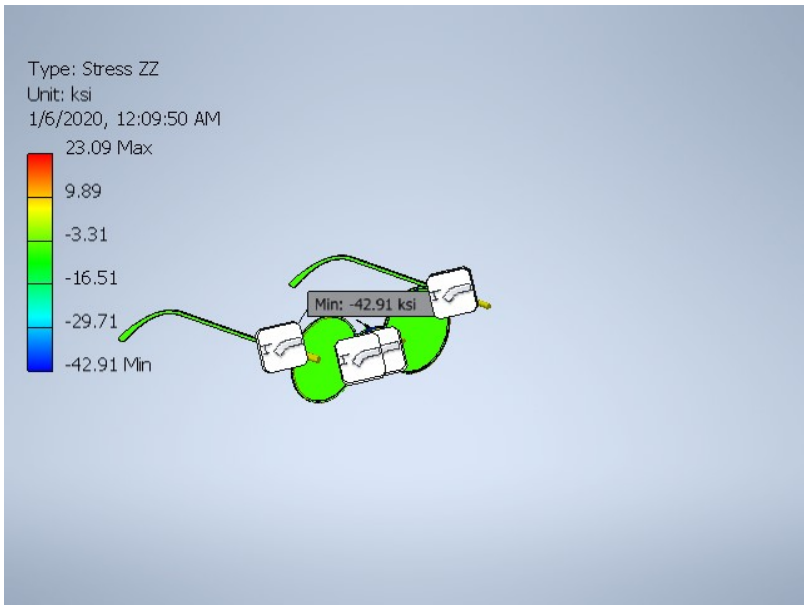
10.62 Max
7.09
3.55
0.02
-3.51
-7.05 Min
Max: 10.62 ksi

⊟ Stress ZZ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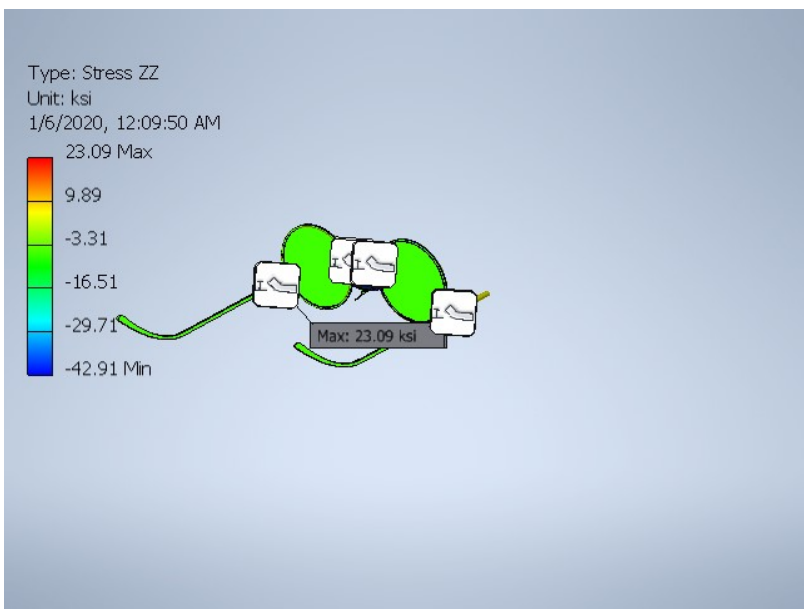
23.09 Max
9.89
-3.31
-16.51
-29.71
-42.91 Min
Min: -42.91 ksi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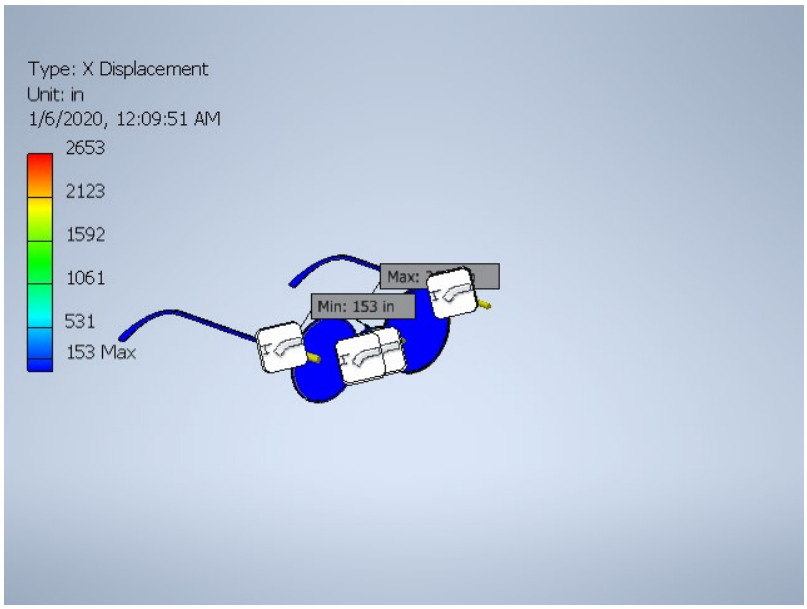
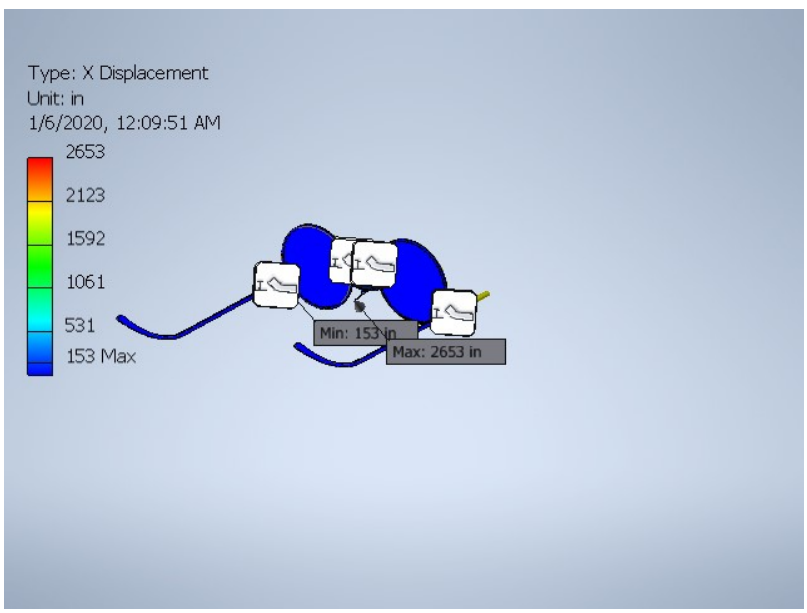
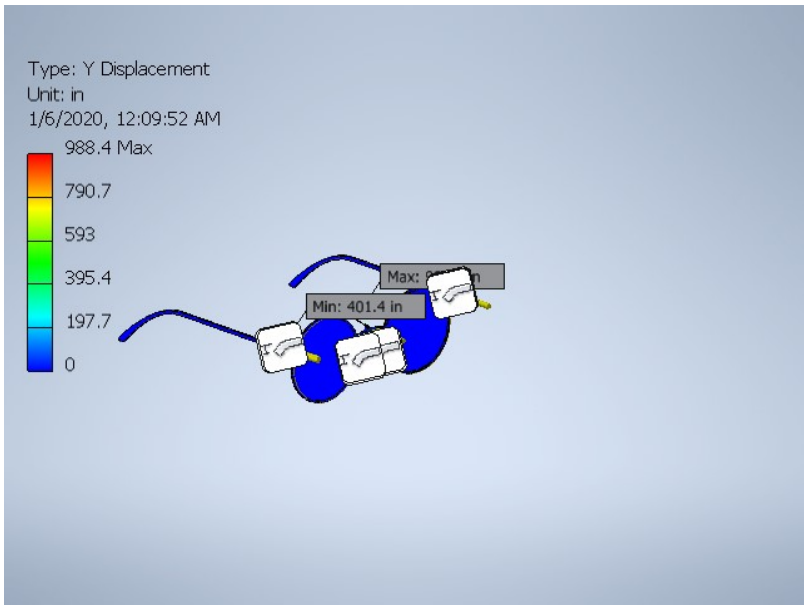
1/6/2020, 12:09:50 AM
23.09 Max
9.89
-3.31
-16.51
-29.71
-42.91 Min
Max: 23.09 ksi

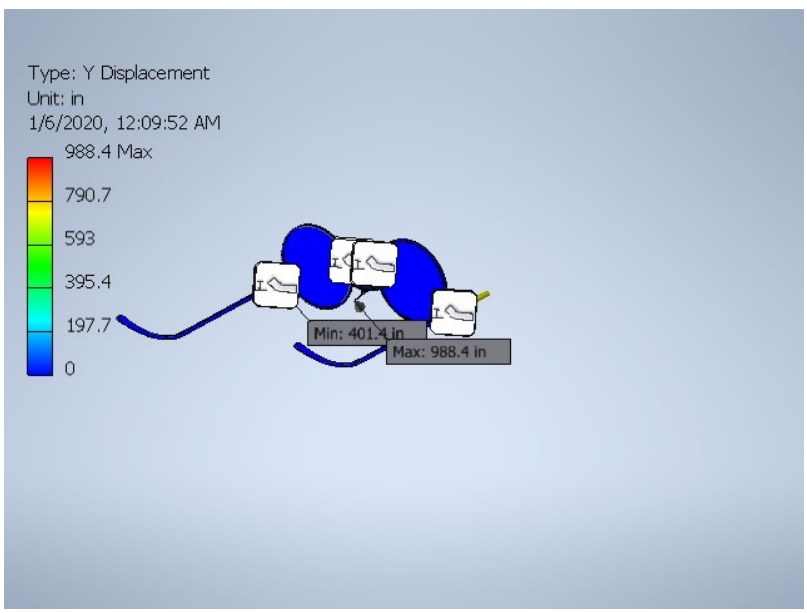
X Displacement



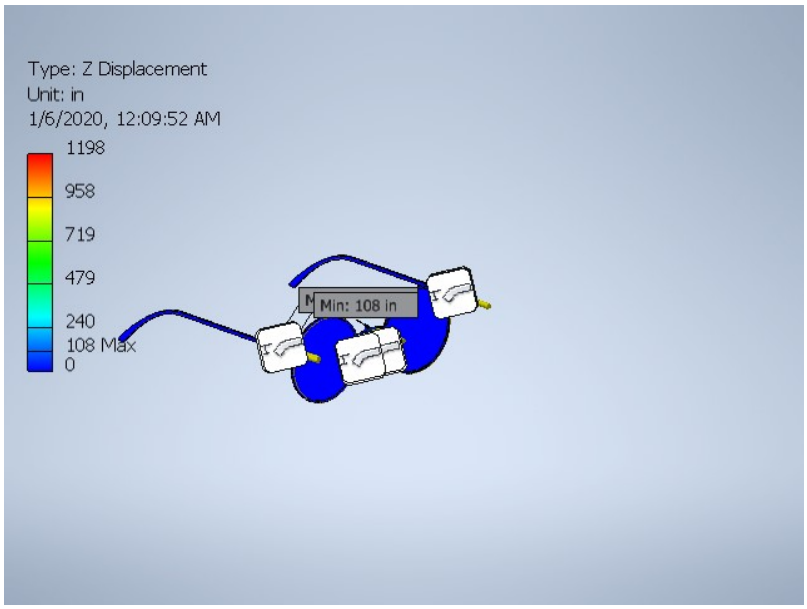


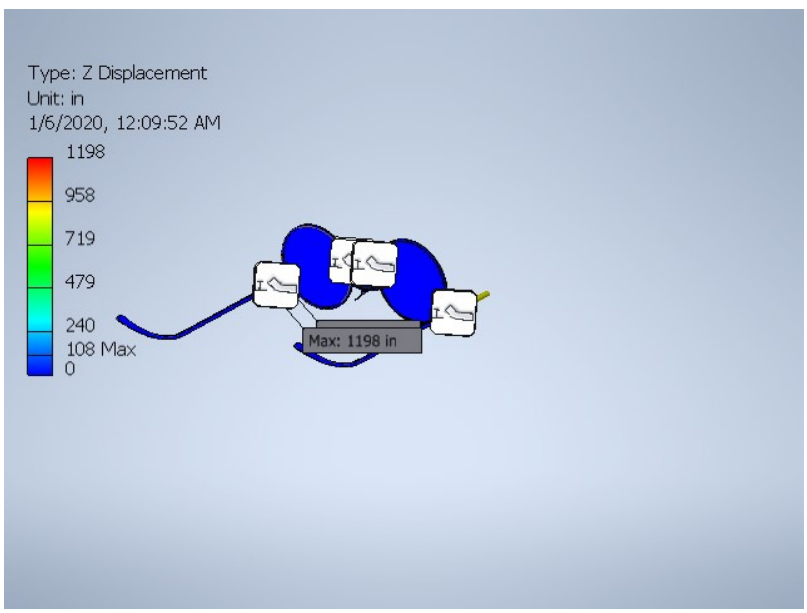
Y Displacement



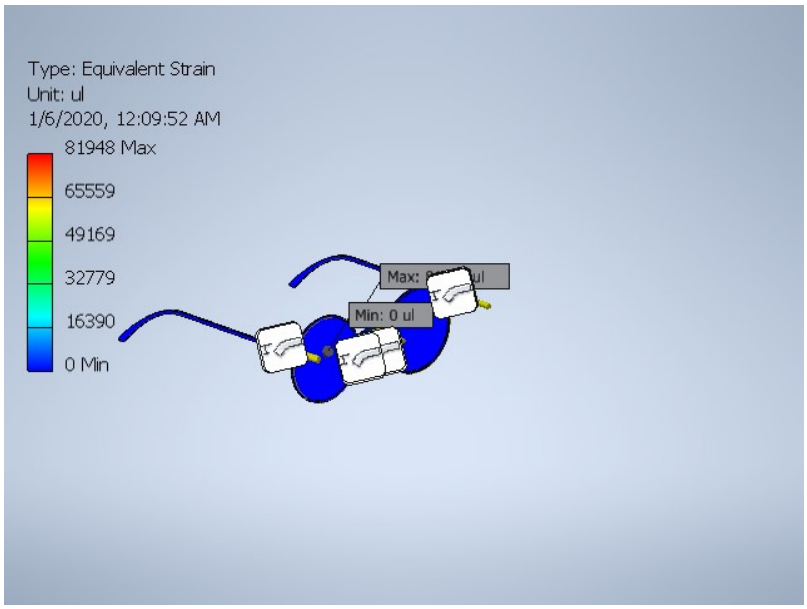


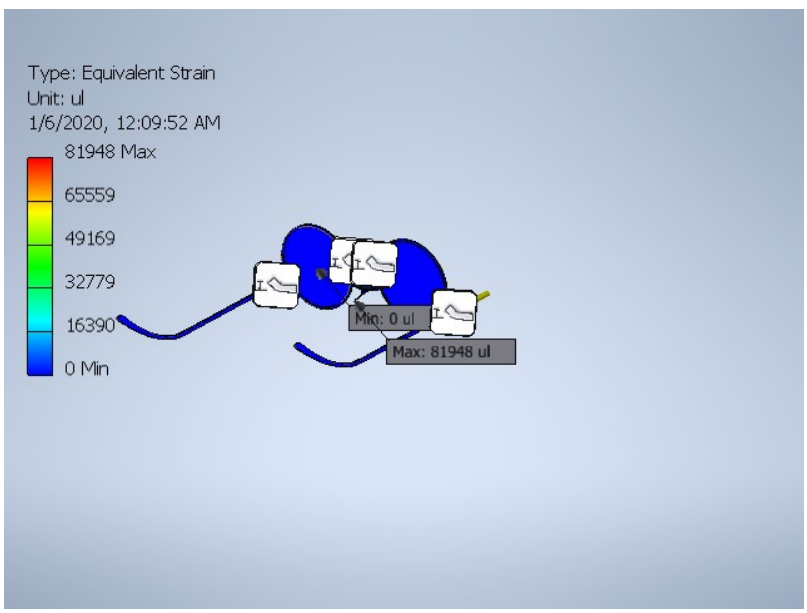
Z Displacement



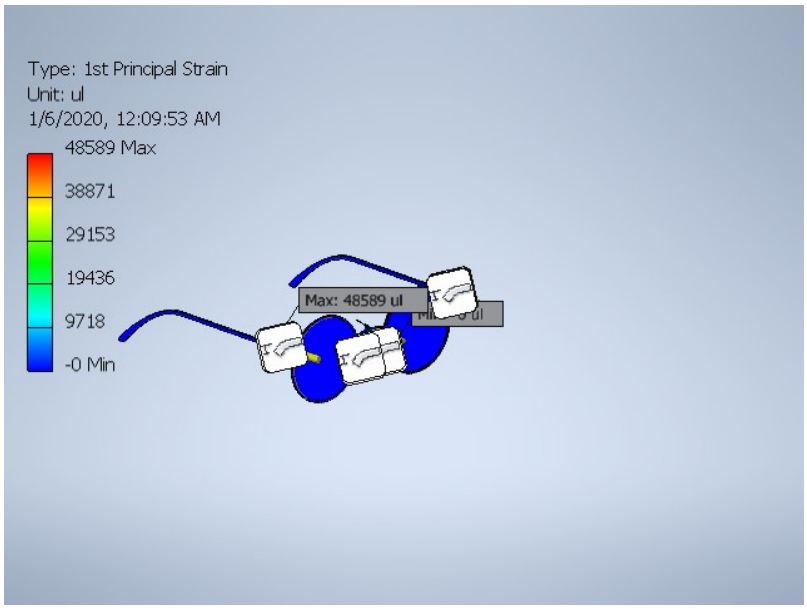


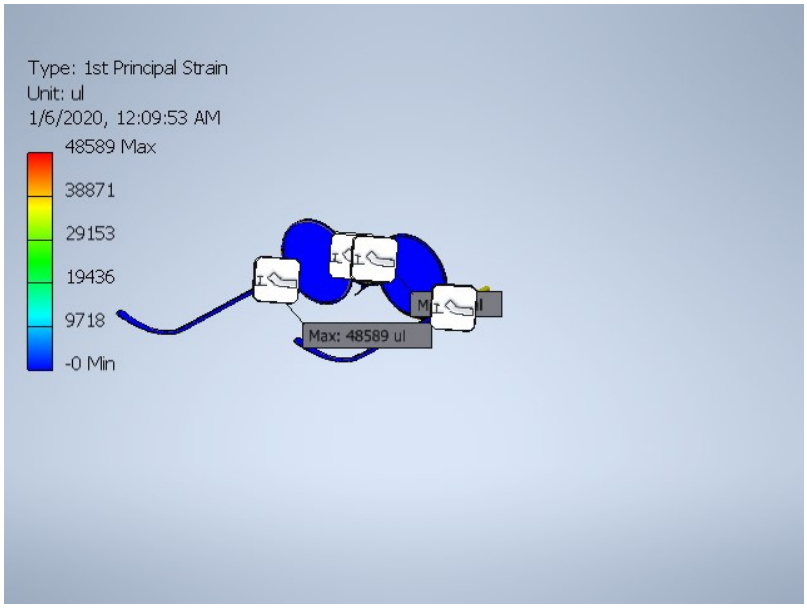
⊟ Equivalent Strain



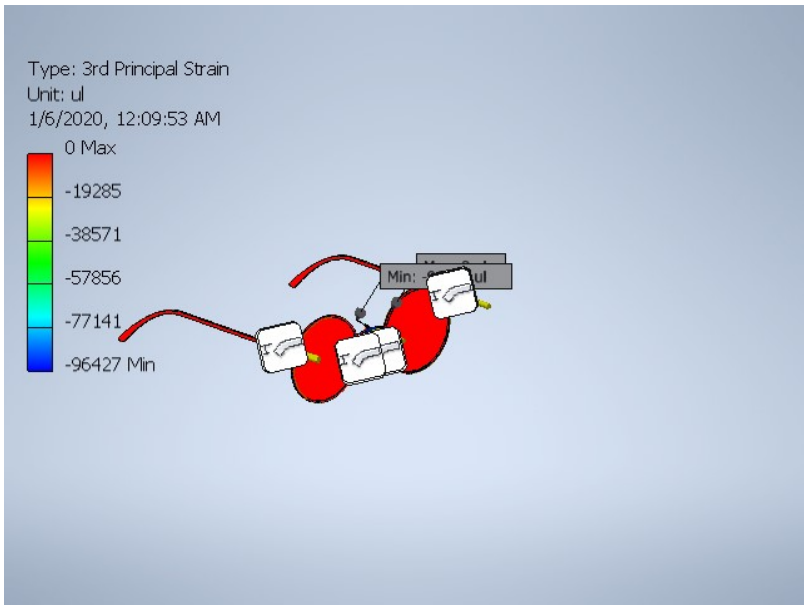


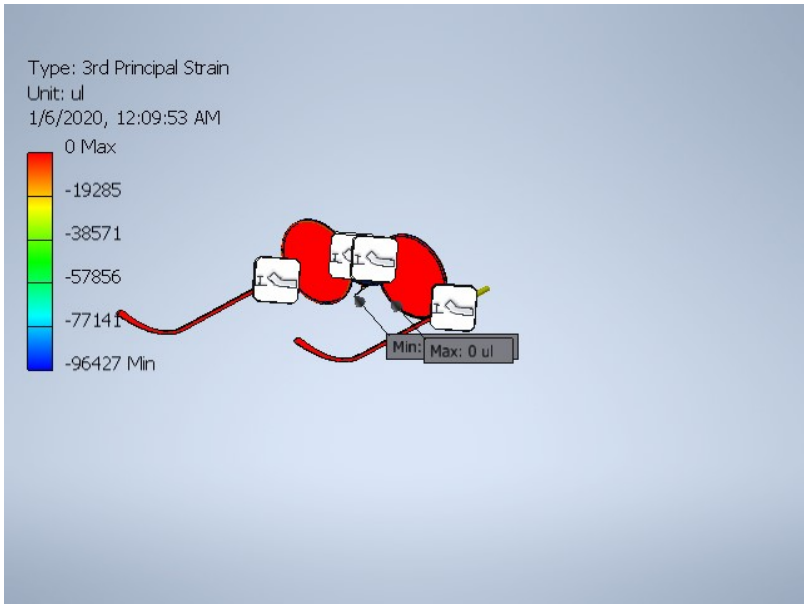
1st Principal Strain



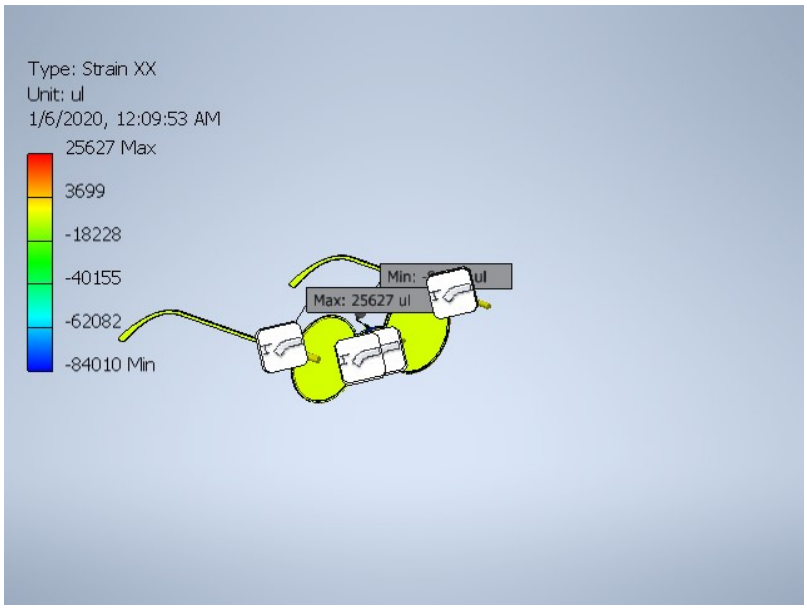


3rd Principal Strain





Strain XX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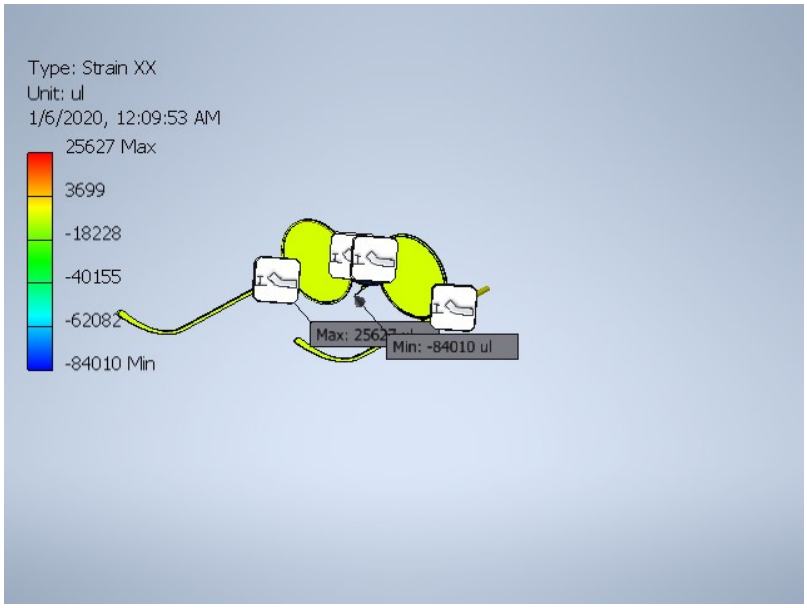
-40155
-62082
-84010 Min
Max: 25627 ul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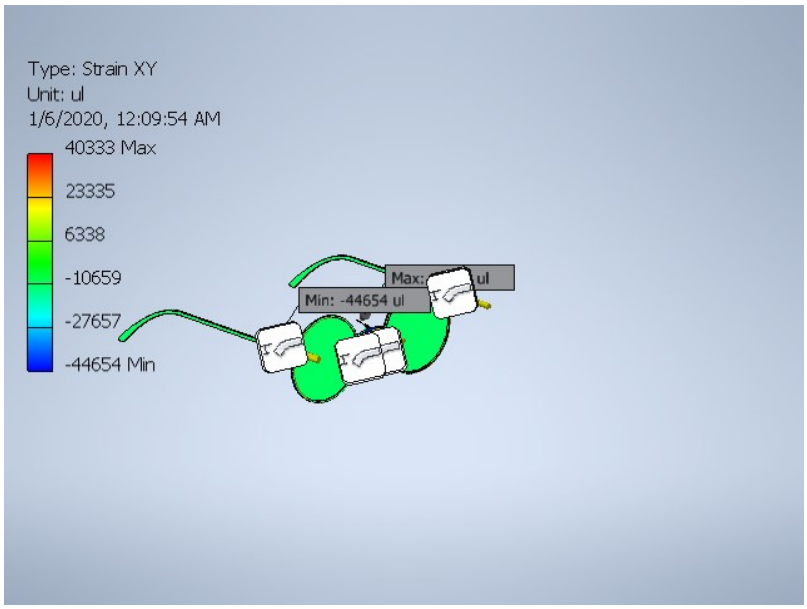
⊟ Strain XY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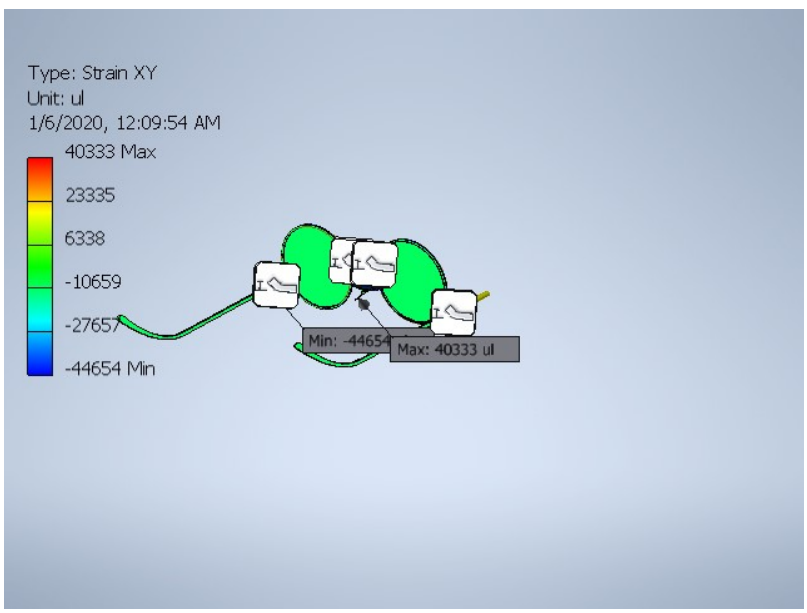
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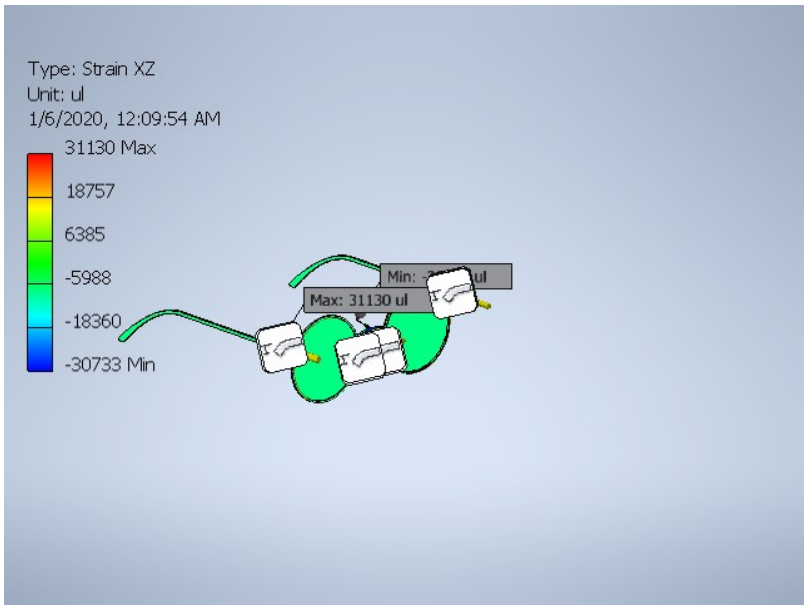
Strain XZ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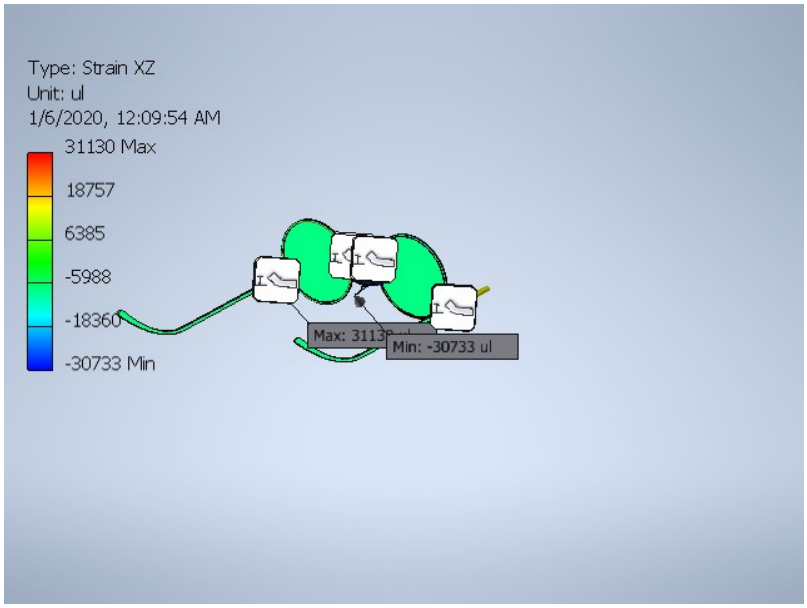
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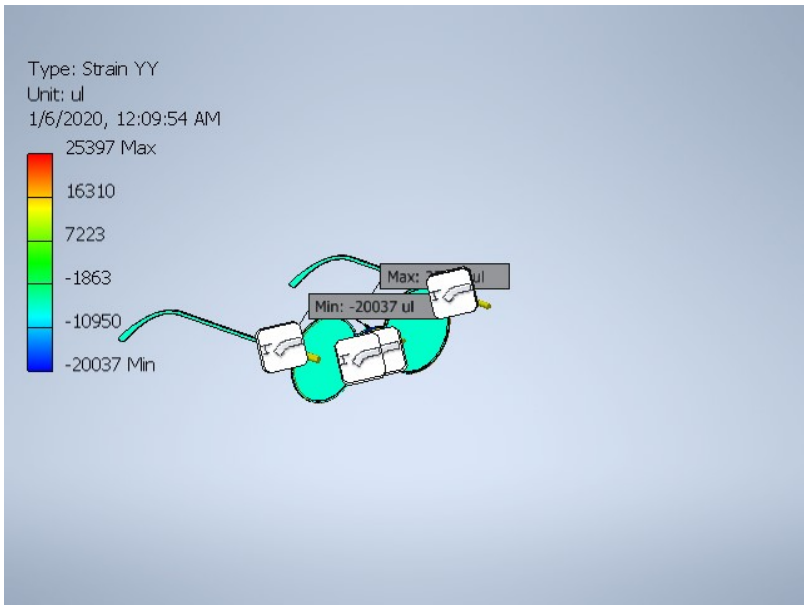
Strain YY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7223
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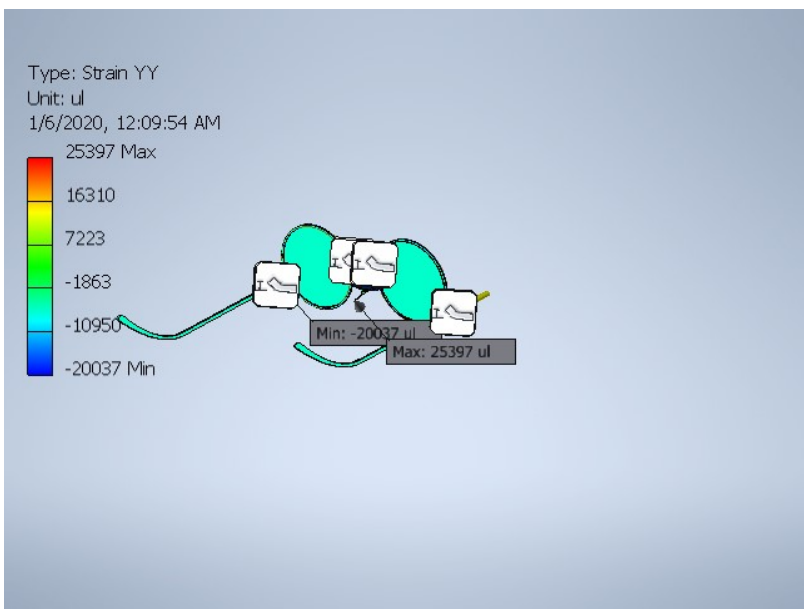
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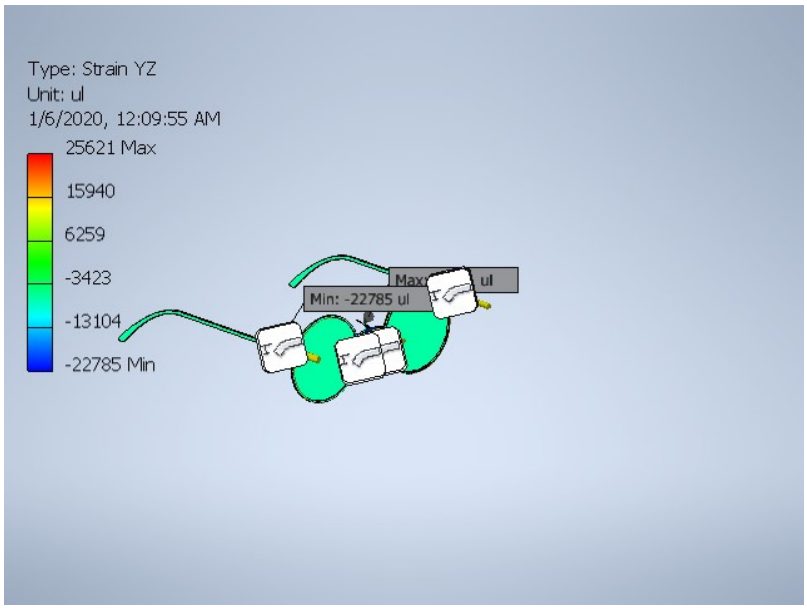
Strain YZ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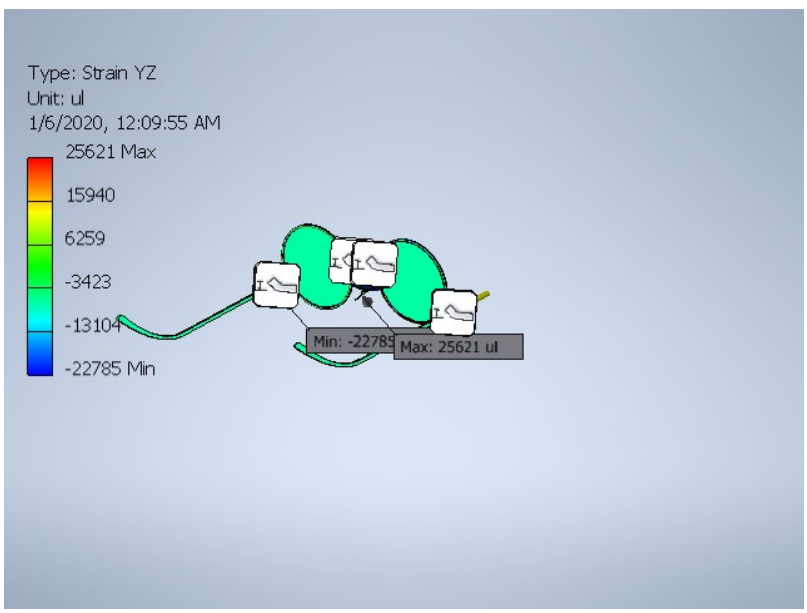
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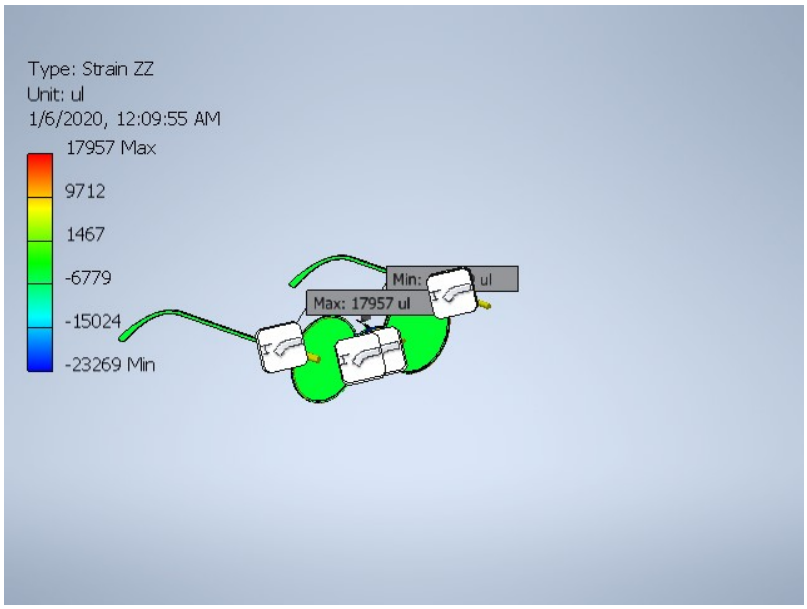
Strain ZZ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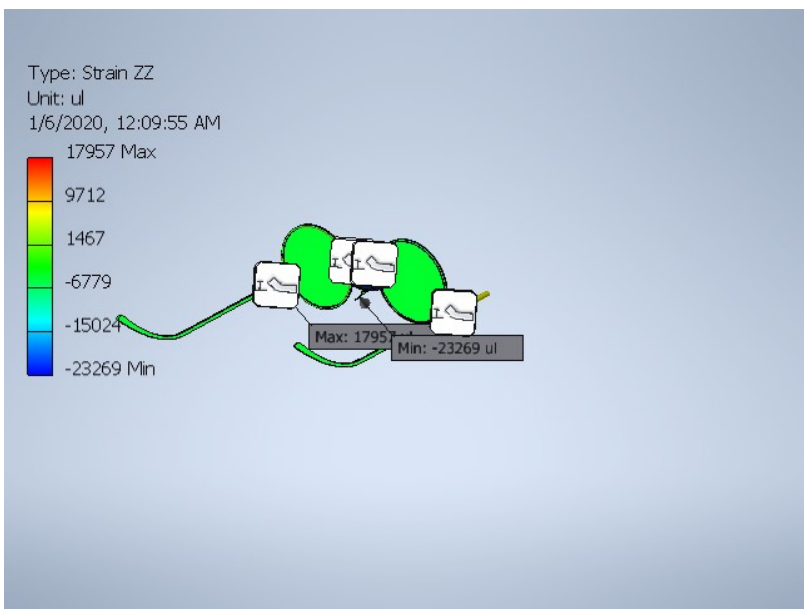
9712
1467
-6779
-15024
-23269 Min
Max: 17957 ul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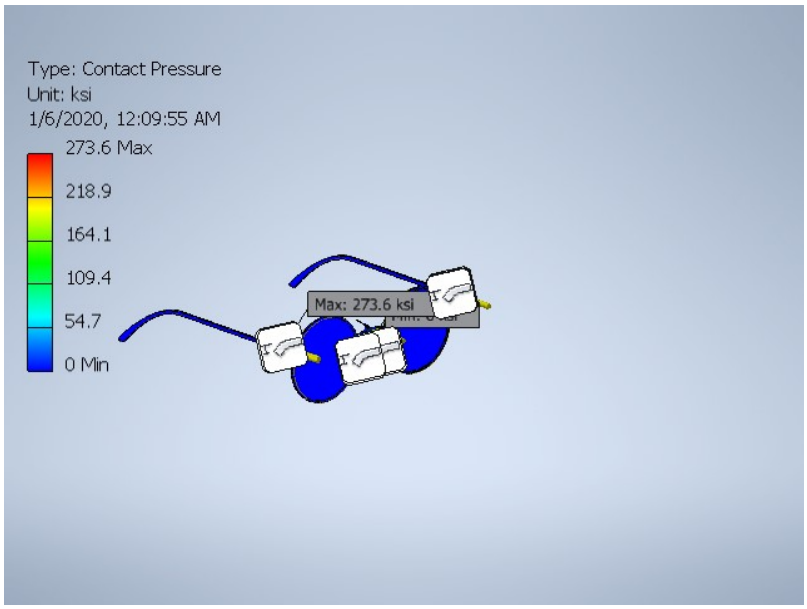
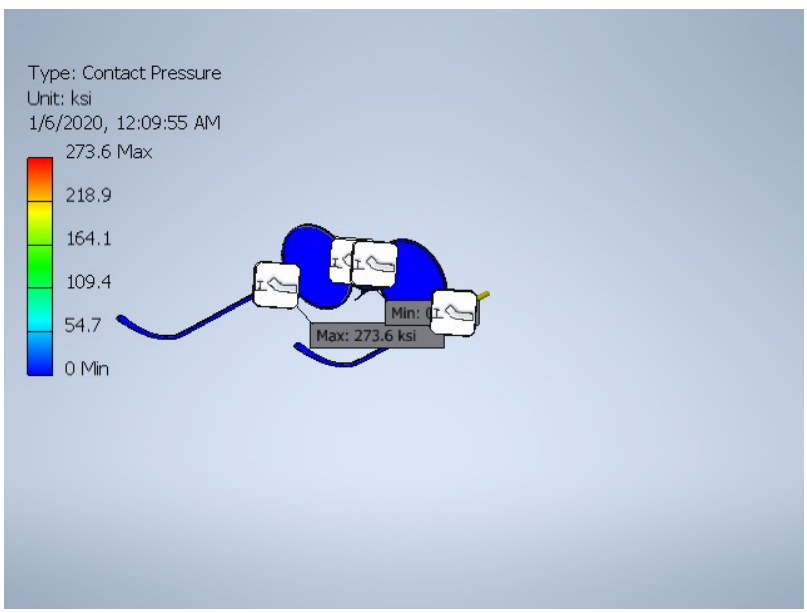
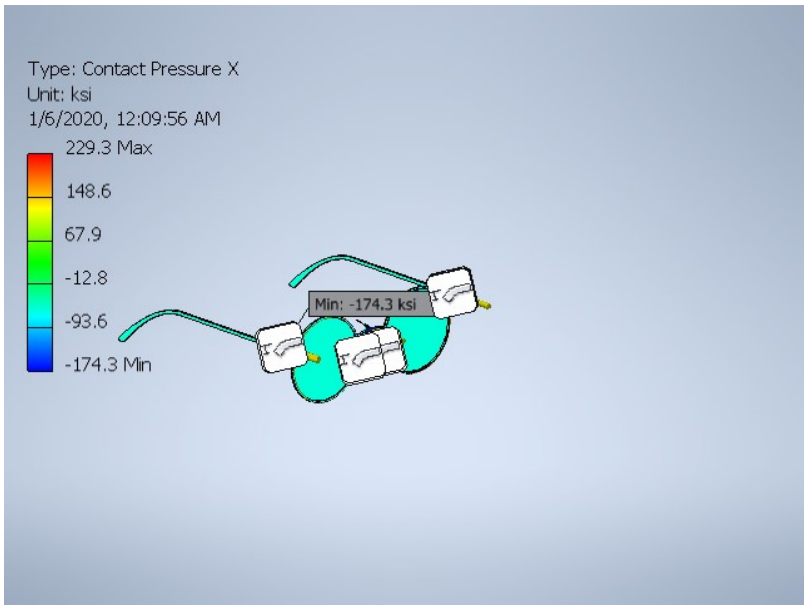
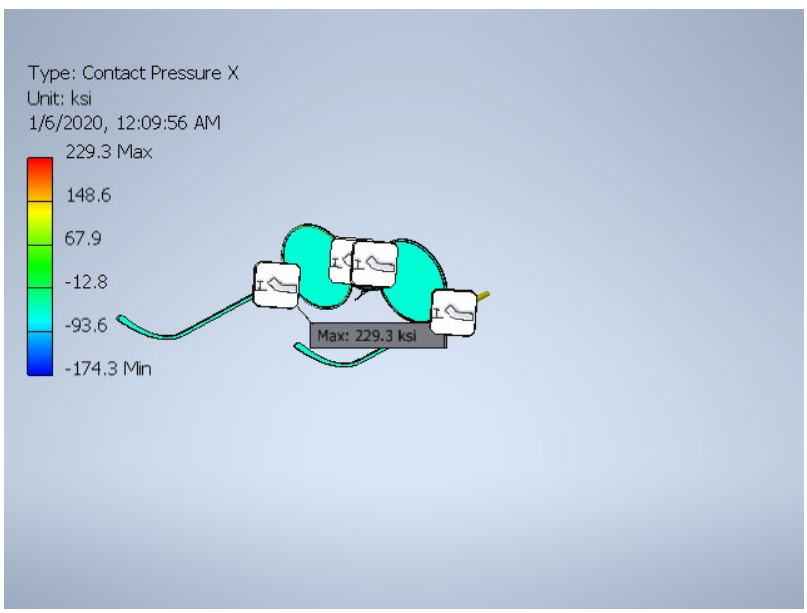
Contact Pressure



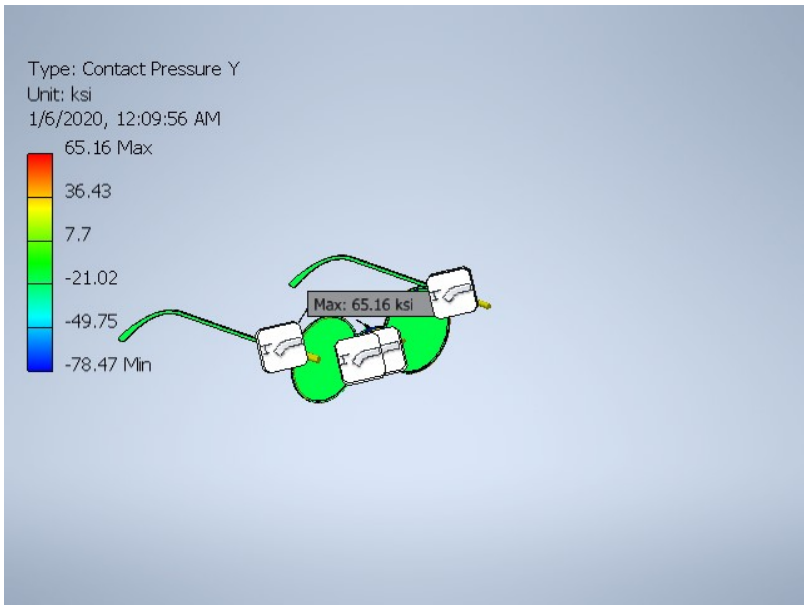


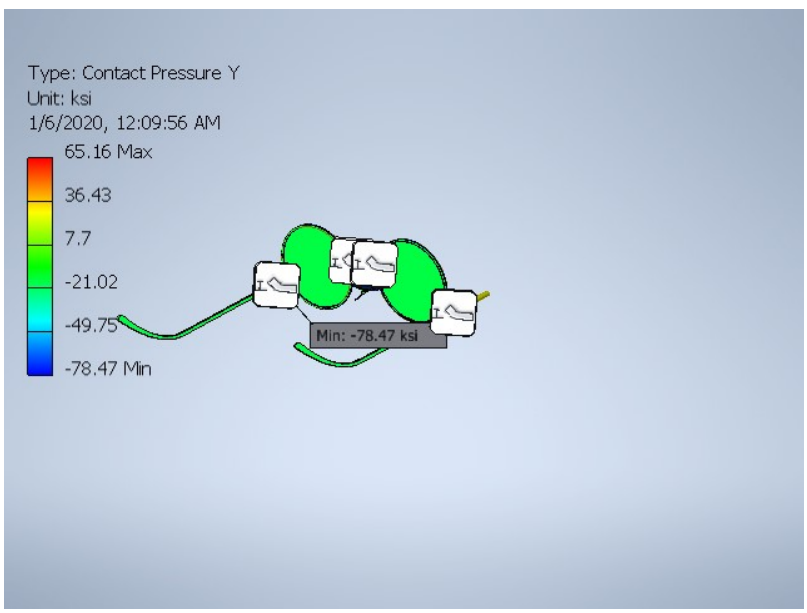
⊟ Contact Pressure X



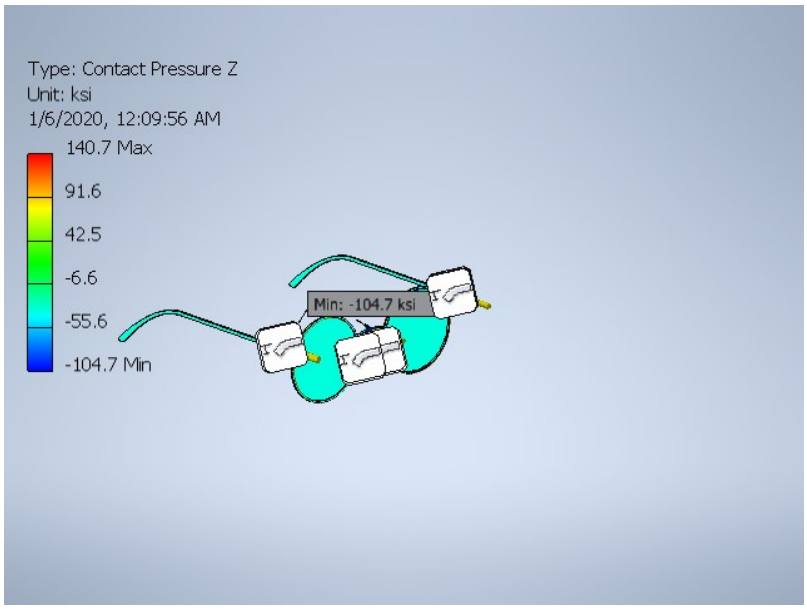


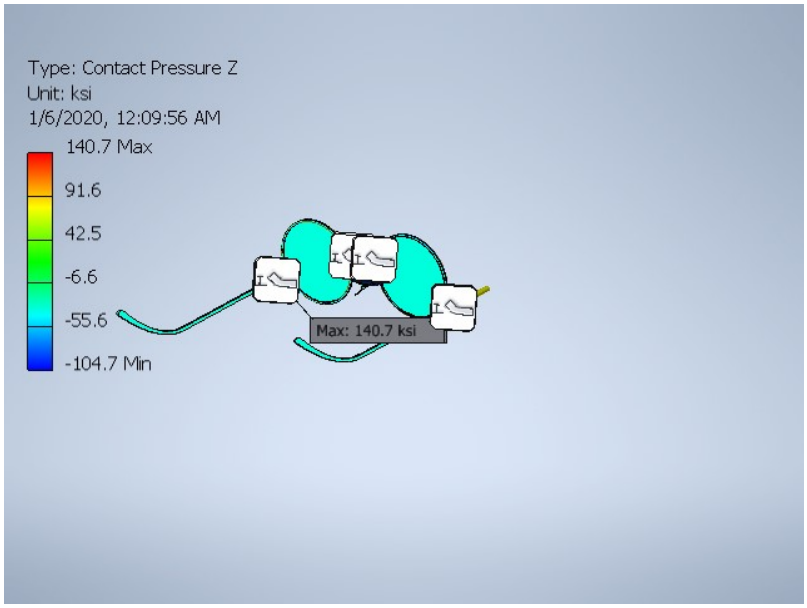
⊟ Contact Pressure Y





⊟ Contact Pressure Z





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